

Institution-Driven Capital Deepening: Industrial Relocation and the Emissions Paradox in Shrinking Cities

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Abstract

Although population constitutes the behavioral subject of consumption-side emissions and a key modulating factor in production-side emissions, empirical evidence suggests that the intuitive notion of passive carbon reduction through economic contraction induced by population outflow may not be reliable. As China enters an era of negative population growth, this issue acquires broader research significance. Using panel data from 297 prefecture-level cities in China over the period 2003–2024, this paper empirically tests this logic. The results indicate that the total carbon emissions of net population-outflow cities are significantly lower than those of net inflow cities solely due to a selection effect, while population outflow per se exerts no clear causal impact on emission trajectories. Mechanism analysis reveals a counter-intuitive pathway: instead of undergoing deindustrialization, cities with population outflow have achieved and sustained industrial relocation—characterized by higher fixed-asset investment growth rates and investment ratios. This process is underpinned by a capital-substitution dynamic driven by institutionally induced capital deepening, shaped jointly by fiscal transfers, local promotion incentives, and strategic investments by state-owned enterprises, thereby maintaining industrial carbon emissions above the market-clearing equilibrium. The findings highlight the distinctive role and critical ecological niche of institutional factors in China’s economic functioning, challenge the mechanistic narrative that population outflow inevitably leads to deindustrialization, and provide empirical evidence and reference for coordinating the spatial reconfiguration of population with the differentiated implementation of the “dual-carbon” strategy.

Keywords: population outflow; shrinking cities; industrial relocation; capital share; green transition

1 Introduction

Population migration and climate change are two core structural transformations of the global economy and society in the twenty-first century, and their interaction has become a frontier research area in environmental and urban economics. Over the past two decades, China has completed the largest spatial restructuring of urbanization in human history, with labor continuously agglomerating from inland cities in the central and western regions toward the eastern coast. In the sample of this paper, 66 percent of the city-year observations from 2003 to 2024 are net population-outflow observations, and this share rose steadily from 60 percent in 2003 to 72 percent in 2024; urban shrinkage has become a normalized phenomenon in China’s regional economic development. The spatial reallocation of population has profoundly reshaped the regional economic geography. Cities, as the core actors in carbon emissions, contribute more than 70 percent of the country’s total emissions (Wu, 2014; Zhang, 2014) and are the key carriers for implementing the dual-carbon goals; yet the dynamic effects and mechanisms of population outflow on urban carbon emissions have not received systematic empirical treatment. Against this

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background, precisely identifying the differentiated effects of population flows on urban carbon emissions and clarifying the evolution of emissions under population shrinkage carry important theoretical value and policy significance.

The classical IPAT framework, the Kaya identity, and the STIRPAT model (York et al., 2003) establish the basic link between population and carbon emissions, forming the mainstream theoretical intuition that population outflow leads to passive carbon reduction through economic contraction. The traditional view holds that population outflow reduces the urban labor supply, shrinks market demand, and cools production and consumption activity, thereby pushing carbon emissions down naturally (Ehrlich and Holdren, 1971; Kaya and Yokobori, 1997). Existing quantitative research in China also supports this perception: demographic change has a non-negligible effect on carbon emissions (Zhu and Peng, 2012; Fan et al., 2021), and the simulations of Zhao (2025) further show that demographic change could reduce cumulative carbon emissions by 3.45 billion tons before 2060, buying an eight-to-ten-year window for China’s carbon neutrality goal. This conclusion reinforces the perception of a positive synergy between population shrinkage and carbon reduction and implies the policy corollary that population policy can assist climate governance.

However, the core limitation of the existing literature is that it takes for granted that population outflow is necessarily accompanied by a proportional contraction of economic activity. This assumption is reasonable under fully market-based conditions, but empirical tests adapted to China’s institutional context remain scarce. China’s distinctive fiscal system, local governance model, and industrial relocation pattern may break the market-based matching relationship among labor, capital, and carbon emissions: if cities with population outflow simultaneously experience capital inflows, industrial relocation, and capacity expansion, the automatic emission-reduction mechanism of population outflow fails completely. Whether population shrinkage necessarily brings carbon reduction is therefore not a purely theoretical question but an empirical proposition that requires micro data to test, and this is the core entry point of this paper.

Accordingly, using panel data for Chinese prefectural-level cities from 2003 to 2024, this paper systematically examines the effects and mechanisms of population outflow on urban carbon emissions. The research proceeds as follows. First, it compares emissions between net-inflow and net-outflow cities along both the level and the growth-rate dimensions, and uses a two-way fixed-effects model to separate the selection effect from the causal effect, identifying the true causal relationship. Second, from the perspective of dynamic industrial restructuring, it explains the structural reasons why emissions in population-outflow cities fail to converge. Third, focusing on fixed-asset investment behavior, it verifies the counter-market feature of “labor outflow with capital agglomeration” in shrinking cities. Fourth, from the perspective of institutional economics, it constructs an analytical framework of institution-driven capital deepening to explain the logic of this peculiar pattern.

The remainder of this paper is organized as follows. Section 2 presents the stylized facts on urban carbon emissions and population flows over the sample period, distinguishing the selection effect from the causal effect. Section 3 builds a simple analytical model and derives the critical condition under which population outflow reduces carbon emissions. Section 4 sets out the sample, variables, and econometric model. Section 5 presents the baseline regressions and the mechanism tests. Section 6 concludes, draws out policy implications, and discusses limitations and directions for future research.

2 Stylized Facts

To clarify the relationship between population outflow and urban carbon emissions, this paper compares emissions between net-outflow and net-inflow cities along both the level and the growth-rate dimensions, and further identifies the core source of the between-group difference, distinguishing the selection effect of inherent characteristics from the causal effect of population

flows.

Figure 1 plots the scale and growth trends of carbon emissions for the two groups of cities. The total carbon emissions of net-outflow cities are always lower than those of other cities, consistent with the classical expectation. Interestingly, however, the emission growth rate of net-outflow cities is close to that of other cities, and the confidence intervals of the two curves always overlap, implying that the growth rates are not significantly different. This phenomenon directly challenges the mechanical logic that population outflow leads to contraction-driven carbon reduction.

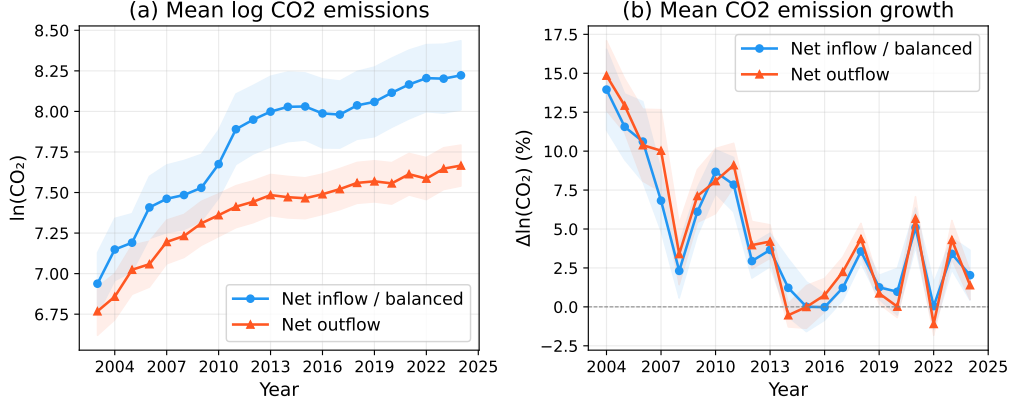


Figure 1: Carbon emissions of net-outflow cities versus other cities.

Table 1 reports more detailed between-group mean-difference tests. In terms of the stock level of emissions, the total carbon emissions of net-outflow cities remain persistently lower than those of net-inflow cities over the sample period. The statistics in Table 1 show that net-outflow cities emit on average about 24.47 million tons of standard coal equivalent per year, while net-inflow cities emit about 37.73 million tons, a between-group difference of 13.26 million tons; the emission scale of net-inflow cities is 35 percent higher than that of net-outflow cities, and the Welch t -test is significant at the 1 percent level. In terms of time-series evolution, however, the log-mean emission curves of the two groups remain roughly parallel, and the between-group gap shows no clear convergence or divergence over the 22 years. This pattern preliminarily confirms the existence of the selection effect: if population outflow had a genuine causal emission-reduction effect, the two emission curves should gradually diverge as outflow intensity accumulates, rather than maintaining a stable fixed gap.

The second panel of Table 1, which compares the dynamic growth rates of emissions, further corroborates this judgment. Over the sample period, the annual emission growth curves of net-outflow and net-inflow cities overlap closely; whether in the mid-2000s phase of rapid emission growth or the post-2010 phase of low growth and stabilization, the growth rates of the two groups do not diverge systematically. The year-by-year t -tests show that the between-group growth differences fail to pass the 5 percent significance test in any year. This indicates that, although net-outflow cities started with lower emission stocks, population flows did not change the dynamic trajectory of urban carbon emissions, and the emission growth paths of the two groups are not statistically different.

Next we directly test the correlation between population outflow and carbon reduction. Figure 2 shows the correlation between outflow intensity and carbon emissions in the full sample: the goodness of fit is extremely low, and no clear trend can be extracted from the pooled scatter. Figure 3 provides the scatter after controlling for city and year fixed effects: the partial regression coefficient of outflow intensity does not reach the 5 percent significance level (for the emission level, $t = 1.68$ and $p = 0.093$), and the fitted line is approximately horizontal, indicating that once inherent city characteristics and time trends are removed, population outflow has no robust

Table 1: Between-group difference tests for carbon emissions

Variable	Net inflow		Net outflow		Difference	Significance	Effect size
	N	Mean	N	Mean			
Emission level (Mtce)	2278	3773	4256	2447	−1326	***	−0.490
$\ln(\text{CO}_2)$	2278	7.744	4256	7.398	−0.346	***	−0.337
$\Delta \ln(\text{CO}_2)$	2151	0.050	4086	0.045	−0.005	†	−0.043
<i>Year-by-year $\Delta \ln(\text{CO}_2)$</i>							
2004	130	13.95%	167	14.85%	+0.90%	†	
2005	149	11.56%	148	12.92%	+1.36%		
2006	131	10.61%	166	10.38%	−0.23%		
2007	117	6.82%	180	10.03%	+3.21%		
2008	115	2.31%	182	3.39%	+1.08%		
2009	118	6.11%	179	7.12%	+1.01%	†	
2010	113	8.67%	184	8.08%	−0.59%		
2011	96	7.85%	201	9.10%	+1.25%		
2012	94	2.94%	203	3.97%	+1.02%		
2013	92	3.66%	205	4.20%	+0.54%		
2014	92	1.22%	205	−0.54%	−1.76%		
2015	94	0.01%	203	−0.00%	−0.01%		
2016	95	−0.02%	202	0.75%	+0.77%		
2017	95	1.22%	202	2.25%	+1.03%		
2018	93	3.56%	204	4.36%	+0.80%		
2019	91	1.26%	206	0.87%	−0.39%		
2020	88	0.97%	209	0.02%	−0.96%		
2021	88	5.09%	209	5.67%	+0.58%		
2022	88	0.05%	209	−1.10%	−1.14%		
2023	87	3.40%	210	4.30%	+0.90%		
2024	85	2.03%	212	1.39%	−0.64%		

Notes: *t*-tests use the Welch correction; effect size: Cohen's *d*. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, † $p < 0.15$.

statistical association with urban carbon emissions.

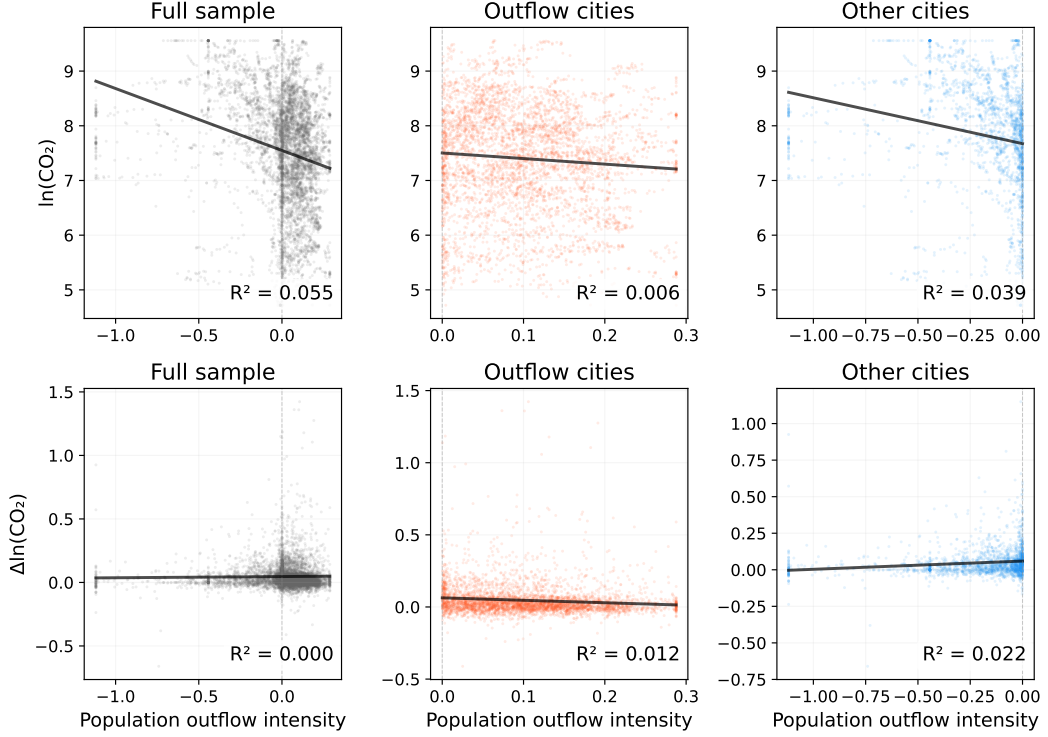


Figure 2: Mixed scatter of population outflow and urban carbon emissions.

In sum, the stylized facts yield a preliminary conclusion: there is no direct causal relationship between urban carbon emissions and population outflow. The stock difference between the two groups of cities is essentially an ex-ante selection effect arising from different initial endowments—net-outflow cities are inherently low-emission-endowment cities, rather than population outflow delivering emission reductions. This conclusion provides the empirical foundation for the theoretical modeling and mechanism tests that follow.

3 Theoretical Analysis

Based on the stylized facts above, this section builds a simple analytical model to derive the critical condition under which population outflow leads to carbon reduction. The classical Kaya identity decomposes total carbon emissions (C) into the product of population size (P), per-capita output (y), and emission intensity (τ):

$$C = P \times y \times \tau \quad (1)$$

Taking logarithms and differentiating, the growth rate of carbon emissions can be decomposed into the sum of the population growth rate, the per-capita output growth rate, and the emission intensity growth rate:

$$\frac{\dot{C}}{C} = \frac{\dot{P}}{P} + \frac{\dot{y}}{y} + \frac{\dot{\tau}}{\tau} \quad (2)$$

The core assumption of the traditional literature is that per-capita output y and emission intensity τ are exogenously fixed, from which the linear conclusion that a decline in population size reduces carbon emissions follows directly. This assumption, however, ignores the endogeneity of factor substitution and capital accumulation, and it fails in settings where factor mobility is restricted in the market.

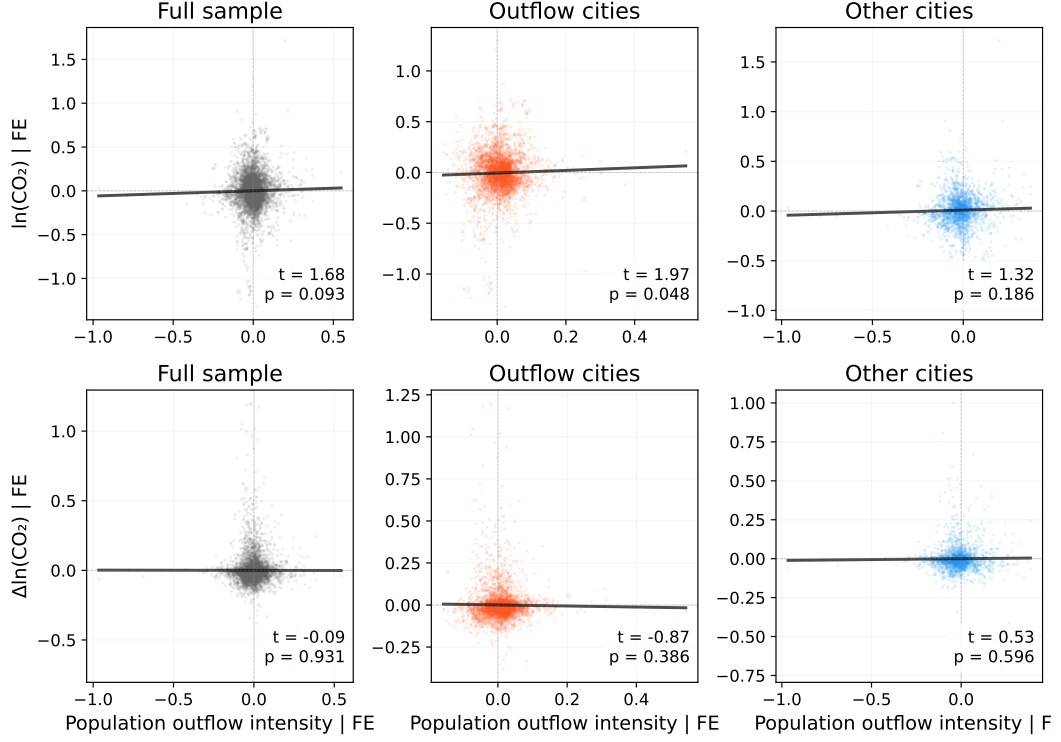


Figure 3: Scatter after controlling for city and year fixed effects.

This paper introduces a Cobb–Douglas production function to characterize urban production:

$$Y = AK^\alpha L^{1-\alpha} \quad (3)$$

where Y is total output, A is total factor productivity, K is the capital stock, L is the labor force, and α is the capital output elasticity. Defining per-capita capital $k = K/L$, per-capita output can be written as:

$$y = \frac{Y}{L} = Ak^\alpha \quad (4)$$

Taking logarithms and differentiating:

$$\frac{\dot{y}}{y} = \frac{\dot{A}}{A} + \alpha \frac{\dot{k}}{k} \quad (5)$$

where $\dot{k}/k = \dot{K}/K - \dot{L}/L$. Substituting into the emission growth equation and assuming that short-run technological progress and emission intensity are unchanged ($\dot{A}/A \approx 0$ and $\dot{\tau}/\tau = 0$), we have:

$$\frac{\dot{C}}{C} = \frac{\dot{P}}{P} + \alpha \left(\frac{\dot{K}}{K} - \frac{\dot{L}}{L} \right) \quad (6)$$

Noting that in the short run labor can be taken as proportional to population ($L \propto P$), equation (6) can be rearranged into the carbon emission growth formula:

$$\frac{\dot{C}}{C} = \alpha \frac{\dot{K}}{K} + (1 - \alpha) \frac{\dot{P}}{P} \quad (7)$$

Based on the above formula, the critical condition for population outflow to achieve carbon reduction can be identified. To achieve a decline in carbon emissions, that is, $\dot{C}/C < 0$, the capital growth constraint must be satisfied:

$$\frac{\dot{K}}{K} < \frac{\alpha - 1}{\alpha} \cdot \frac{\dot{P}}{P} \quad (8)$$

Since $\dot{P}/P < 0$ under population outflow and $\alpha \in (0, 1)$, the right-hand side of equation (8) is positive. The condition therefore means that population outflow can bring carbon reduction only when the capital growth rate falls below a certain positive threshold. It follows that population outflow does not necessarily lead to carbon reduction: if capital accumulation is fast enough, it can completely offset or even exceed the emission-reduction effect of population outflow, pushing carbon emissions persistently upward (Du and Li, 2024; Lin and Zhu, 2021).

The operational features of China’s urban economy—in which the resource-allocation inefficiency caused by factor-market distortions is particularly prominent (Lin and Chen, 2018)—happen to break the market-based emission-reduction critical condition above, forming a disequilibrium pattern in which capital growth is systematically too high. The existing literature provides theoretical support from four dimensions: industrial relocation, the fiscal system, governance incentives, and state-owned-enterprise investment. First, industrial relocation: since the mid-2000s, eastern coastal manufacturing has migrated on a large scale to the inland central and western regions, and net population-outflow cities there have become the core carriers of industrial relocation, sustaining the industrialization process (Zhang and Liu, 2013; Fan, 2019). Second, the fiscal system: central fiscal transfers have effectively improved the public spending and investment capacity of central and western cities, detaching their investment behavior from local economic endowments and market supply-and-demand constraints (Fan and Zhang, 2010; Xi et al., 2017). Third, governance incentives: the promotion-tournament mechanism for local officials generates strong investment impulses, and local governments generally rely on fixed-asset investment to drive economic growth, forming a persistent investment preference (Zhou, 2007; Xu et al., 2007). Fourth, the behavior of economic agents: state-owned enterprises, relying on national strategic deployment, undertake non-market investments in infrastructure, energy, and other fields, and are heavily concentrated in shrinking cities in the central and western regions, further strengthening regional capital accumulation. Together, these institutional factors shape a capital-deepening pattern distinct from the fully market-based equilibrium, keep the capital growth of population-outflow cities persistently above the market equilibrium level, break the emission-reduction critical condition of population outflow, and finally produce the green paradox of “population outflow without falling emissions.” At the same time, the green Solow model and the theory of sectoral resource reallocation corroborate this logic: when capital accumulation is detached from market signals, the decline in environmental pressure lags significantly behind the pace of population shrinkage (Stockman, 1981; Brock and Taylor, 2010).

4 Research Design

4.1 Data and Variables

This paper uses a balanced panel of 297 prefectural-level cities in China from 2003 to 2024. Urban carbon emission data come from the China Emission Accounts and Datasets (CEADs), which are estimated on the basis of energy balance sheets and unified emission factors and are highly accurate and authoritative. Population data (registered population and resident population) and economic development data (GDP, fixed-asset investment, and the value-added shares of the secondary and tertiary industries) are compiled from the statistical yearbooks of each prefectural-level city and the national economic and social development statistical bulletins. The policy variables for the low-carbon city pilot and the carbon emission trading pilot are manually constructed from the batch lists published by the National Development and Reform Commission. To avoid the interference of extreme values in the regressions, all continuous variables are winsorized at the 1st and 99th percentiles.

To precisely distinguish stock differences from dynamic evolution differences, this paper measures carbon emissions along two dimensions. The first is the stock level of emissions, measured by the log of total carbon emissions, $\ln(\text{CO}_2)$; the second is the dynamic growth rate of

emissions, measured by the first difference of the log, $\Delta \ln(\text{CO}_2)$, which enables the effective identification of the selection effect versus the causal effect. Following the mainstream measurement of shrinking cities, the intensity of population outflow is measured by the continuous indicator (registered population – resident population)/registered population. A positive value indicates net population outflow, with larger values indicating stronger outflow intensity; a negative value indicates net population inflow.

4.2 Model Specification

To test the baseline effect of population outflow on urban carbon emissions, this paper constructs the following two-way fixed-effects model:

$$\ln \text{CO}_{2,it} = \beta \cdot \text{Outflow}_{it} + \gamma' X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (9)$$

where i denotes the city and t the year; $\ln \text{CO}_{2,it}$ is the log of city i 's carbon emission level in year t ; Outflow_{it} is the population outflow intensity; X_{it} is a vector of control variables; μ_i is the city fixed effect, which removes the time-invariant inherent endowment differences across cities; λ_t is the year fixed effect, which controls for nationwide common time trends and macro shocks; and ε_{it} is the random disturbance term.

This paper identifies the selection effect and the causal effect through a stepwise regression strategy: it first estimates a pooled OLS model, then gradually adds city fixed effects, year fixed effects, and control variables, using the changes in the sign and significance of the core coefficient to distinguish spurious cross-sectional correlation from the true causal effect.

5 Empirical Analysis

5.1 Baseline Model

Table 2 reports the stepwise baseline results, clearly presenting the dynamic evolution of the core coefficient. The first column, pooled OLS, shows that the coefficient on outflow intensity is significantly negative at the 1 percent level, initially confirming the traditional “population-outflow reduction” perception. This result, however, only reflects cross-sectional correlation and does not remove the interference of inherent city characteristics. After adding city fixed effects in the second column, the core coefficient reverses fundamentally, turning from negative to positive and significant at the 5 percent level, indicating that the significant negative effect in the pooled OLS is entirely due to cross-sectional endowment differences across cities—that is, net-outflow cities are inherently low-emission cities, rather than population outflow causing emission reduction. After further controlling for year fixed effects in the third column, the coefficient falls sharply and is no longer significant. After the fourth and fifth columns successively add industrial structure, economic level, and policy pilot controls, the core coefficient keeps approaching zero and is completely insignificant. At the same time, the industrial structure coefficient is always significantly positive at the 1 percent level, indicating that the level of industrialization is the core factor driving the growth of urban carbon emissions.

The significant negative correlation in the pooled OLS model is only a spurious cross-sectional association; its essence is an omitted-variable bias caused by the failure to strip out differences in cities' inherent endowments—low-carbon-endowment cities are more prone to population outflow, rather than population outflow causing emission reduction. After stepwise controls for individual fixed effects, time fixed effects, and core control variables, the core coefficient turns from negative to positive and finally approaches zero without any statistical significance, indicating that after removing cities' inherent characteristics, macro time trends, and industrial and policy interference, population outflow has no significant causal emission-reduction effect on urban carbon emissions. This result strongly overturns the mechanical linear

Table 2: Panel regressions of urban carbon emissions on population outflow

	City CO ₂ emissions (log)				
	(1)	(2)	(3)	(4)	(5)
Outflow intensity	-1.133*** (0.273)	0.535** (0.232)	0.091 (0.111)	0.020 (0.105)	0.060 (0.110)
Secondary industry share				0.005*** (0.002)	0.005*** (0.002)
Log per-capita GDP				0.060 (0.045)	0.066 (0.045)
Low-carbon city pilot					0.028 (0.030)
Carbon trading pilot					0.074 (0.062)
City fixed effects	No	Yes	Yes	Yes	Yes
Year fixed effects	No	No	Yes	Yes	Yes
<i>N</i>	6,348	6,348	6,348	6,348	6,348
<i>R</i> ²	0.058	0.893	0.964	0.965	0.966

Notes: City-level cluster-robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

logic that “population shrinkage necessarily reduces emissions” under the traditional IPAT and Kaya frameworks, confirms that the relationship between population and carbon emissions at the level of Chinese cities is completely different from the expectation of fully market-based theory, and explains why shrinking cities in reality do not show the theoretically expected emission decline. It lays a solid empirical foundation for the structural mechanism tests and institutional mechanism analysis below. At the same time, the persistently significant positive coefficient on industrial structure further corroborates that the level of industrialization is the core market factor determining the scale and trajectory of urban carbon emissions, while the marginal explanatory power of population flows is essentially negligible.

5.2 Industrial Relocation

According to the analysis above, the reason the emission trajectory of net-outflow cities remains stable is that they host industrial relocation; this subsection tests this logic by comparing the time-series evolution of industrial structure between the two groups, focusing on the between-group differences in deindustrialization. Figure 4 shows the industrial structure levels of the two groups: the secondary industry share, the tertiary industry share, and the industrial upgrading index of net-outflow cities are all lower than those of net-inflow cities, reflecting the structural difference in initial endowments. The fluctuation curves in Figure 5, however, are highly similar, with overlapping confidence intervals, again indicating the selection effect: the distance between the curves in Figure 4 comes only from initial endowments, and the two groups show no significant difference in the trend of industrial restructuring.

The persistence of the industrial structure relies on sustained capital accumulation. We further verify the capital-deepening feature of population-outflow cities from three dimensions: the growth rate of fixed-asset investment, the investment rate, and economic output. The fixed-asset investment curves in Figure 6 show that the per-capita fixed-asset scale of net-outflow cities is smaller than that of other cities, but the growth rate is higher in most years—consistent with the logic of industrial relocation.

The more detailed time-series comparison in Table 3 shows that from 2004 to 2010, the fixed-asset investment growth rate of net-outflow cities was persistently higher than that of net-inflow cities, passing the 5 percent significance test in most years. In terms of the investment rate, the fixed-asset-investment-to-GDP ratio of net-outflow cities rose steadily from 0.37 in 2003 (0.04

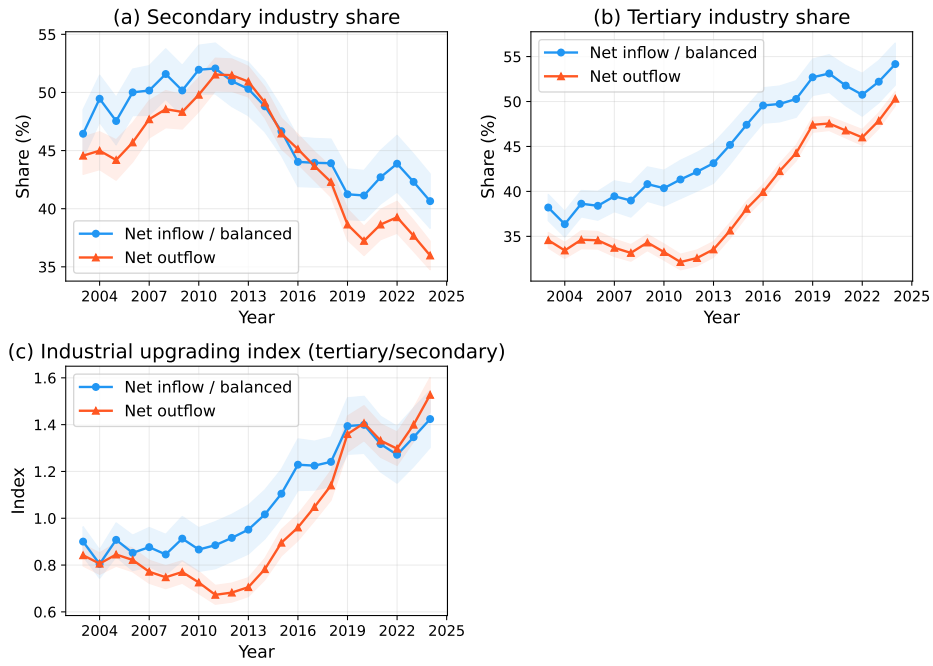


Figure 4: Dynamic comparison of industrial structure between net-outflow cities and other cities.

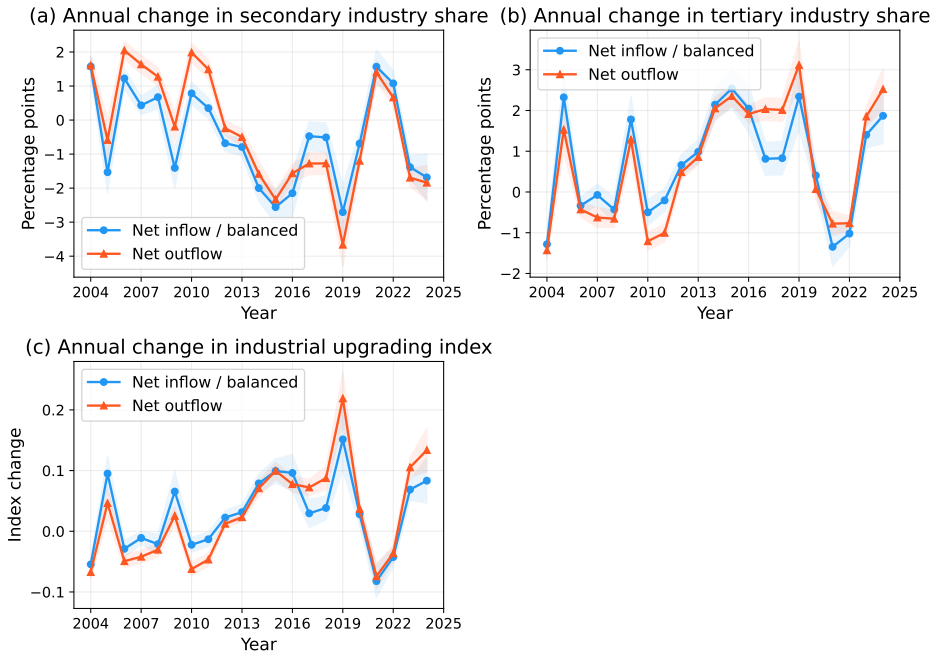


Figure 5: Annual changes in industrial structure: net-outflow cities versus other cities.

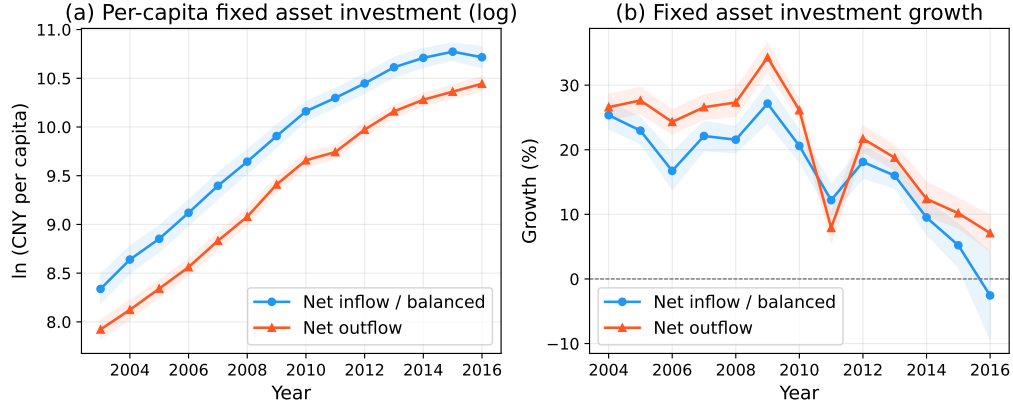


Figure 6: Fixed-asset investment: net-outflow cities versus other cities.

lower than net-inflow cities) to 0.95 in 2016 (0.24 higher than net-inflow cities); the gap widened continuously over the 13 years, showing no sign of convergence.

Table 3: Yearly t -tests of the fixed-asset investment growth rate

Year	Net inflow/balanced	Net outflow	Difference (pp)	t -value	Significance
2004	25.36%	26.59%	+1.23%	-0.83	
2005	22.96%	27.64%	+4.68%	-3.27	**
2006	16.73%	24.31%	+7.58%	-4.37	***
2007	22.13%	26.58%	+4.45%	-2.99	**
2008	21.56%	27.32%	+5.76%	-3.83	***
2009	27.15%	34.29%	+7.14%	-3.79	***
2010	20.59%	26.13%	+5.53%	-3.86	***
2011	12.20%	7.90%	-4.30%	2.56	*
2012	18.11%	21.71%	+3.60%	-2.20	*
2013	15.99%	18.77%	+2.77%	-2.02	*
2014	9.51%	12.39%	+2.87%	-1.54	†
2015	5.21%	10.18%	+4.97%	-2.40	*
2016	-2.56%	7.10%	+9.66%	-2.62	**

Notes: Welch t -tests; difference = net-outflow group mean - net-inflow group mean. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, † $p < 0.15$. After 2017, the National Bureau of Statistics adjusted the statistical caliber of fixed-asset investment, and prefecture-level data are missing.

The regression results in Table 4 further corroborate these features. After controlling for two-way fixed effects, the partial effects of outflow intensity on total fixed-asset investment, the fixed-asset investment rate, and per-capita GDP are 0.485, 0.554, and 0.723, respectively, all significant at the 1 percent level. This indicates that population outflow not only fails to suppress urban investment and economic growth but is actually associated with regional capital accumulation and capacity expansion, forming a counter-market pattern of “labor outflow, capital deepening, and industrial persistence.”

5.3 Policy Interpretation

Based on the empirical results above, this subsection further attempts to explain the internal mechanism of institution-driven capital deepening. In a fully market-based environment, labor outflow would trigger a synchronized outflow of capital, achieving factor matching and market equilibrium; but China’s distinctive institutional system breaks this market-clearing mechanism.

First, the fiscal backstop effect of fiscal transfers. Central vertical fiscal transfers effectively fill the fiscal gaps of population-outflow cities in the central and western regions (Weng, 2026),

Table 4: Panel regressions of capital deepening on population outflow

	Fixed asset investment (log) (1)	FAI / GDP (2)	Real per-capita GDP (log) (3)
Outflow intensity	0.485*** (0.159)	0.554*** (0.132)	0.723*** (0.074)
City fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
<i>N</i>	4,002	4,002	5,857
<i>R</i> ²	0.967	0.724	0.990

Notes: City-level cluster-robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Models (1) and (2) are limited to the 2003–2016 window in which fixed-asset investment data are available.

so that their public investment and industrial supporting investment no longer depend on local economic returns and tax revenue. This forms an independent investment capacity detached from local factor endowments, providing financial support for sustained capital accumulation.

Second, the investment orientation effect of local promotion incentives. Under an official evaluation system centered on economic growth, local governments have strong incentives to expand scale. Facing the growth pressure of population outflow, local officials prefer to stabilize growth by increasing fixed-asset investment, hosting low-end industries, and expanding industrial capacity, persistently strengthening regional capital deepening.

Third, the policy amplification effect of state-owned-enterprise strategic investment. SOE investment relies more on national regional development strategy than on short-term market returns. Population-outflow cities in the central and western regions, as key deployment areas for balanced regional development, host a large amount of SOE investment in infrastructure and the energy industry, further entrenching the non-market capital accumulation pattern.

The institution-driven capital deepening formed by the combination of the three completely offsets the emission-reduction effect of population outflow, keeps the carbon emissions of shrinking cities above the market equilibrium level, and finally produces the green paradox under population shrinkage.

6 Conclusion

Based on panel data for 297 Chinese prefectural-level cities from 2003 to 2024, this paper systematically examines the effects, structural logic, and institutional mechanisms of population outflow on urban carbon emissions, and reconstructs the understanding of the population–emissions relationship in shrinking cities. The findings are as follows.

First, population outflow has no significant emission-reduction effect on urban carbon emissions. The lower emission stock of net-outflow cities is only a sample selection effect—a spurious correlation caused by cities’ initial low-carbon endowments. After controlling for inherent city characteristics and time trends, population outflow has no significant association with either the level or the growth rate of carbon emissions, and the dynamic emission trajectories of the two groups are basically convergent; the linear logic that population shrinkage naturally reduces emissions does not hold at the level of Chinese cities.

Second, shrinking cities exhibit significant “deindustrialization failure” and capital deepening. By hosting industrial relocation from the east, population-outflow cities maintained a stable level of industrialization over the long run, without the industrial contraction and capacity exit predicted by market theory. At the same time, relying on persistently high fixed-asset investment growth and investment rates, they formed a counter-market pattern of “labor outflow, capital agglomeration, and industrial persistence”—the direct reason why emissions did not fall with population outflow.

Third, institution-driven capital deepening is the core mechanism behind the green paradox. The financial support of fiscal transfers, the investment incentives of local official promotion, and the strategic non-market investment of state-owned enterprises jointly push capital accumulation in population-outflow cities beyond the market equilibrium constraint, keep capital growth persistently above the market level, and offset the emission-reduction effect of population outflow, forming the green paradox of shrinking cities with Chinese characteristics.

The conclusions of this paper provide new ideas for the differentiated implementation of the dual-carbon strategy and the coordinated optimization of population and environmental policies.

First, discard the inertial policy perception that “population shrinkage automatically reduces emissions.” Population flows are not the core driver of urban carbon emissions, and population outflow cannot be used as a passive tool of climate governance. The emission-reduction campaign in central and western shrinking cities cannot rely on the natural loss of population; it must formulate differentiated emission-reduction plans based on their own industrial and investment characteristics.

Second, optimize regional investment and fiscal incentive institutions to resolve the institutional green paradox. Carbon emission performance indicators should be incorporated into the fiscal transfer system to constrain disorderly and inefficient industrial investment; the evaluation system for local officials should be improved, weakening the weight of pure investment scale and economic growth and strengthening the guiding role of green, low-carbon, and high-quality development indicators, so as to curb non-market high-carbon capital deepening at the institutional source.

Third, implement differentiated low-carbon transition strategies for shrinking cities. The emission constraints, industrial structure, and investment patterns of central and western population-outflow cities are fundamentally different from those of eastern cities, so eastern emission-reduction experience cannot be copied. Instead, it is necessary to promote the green upgrading of relocated industries, strictly control high-energy-consumption fixed-asset investment, and improve capital utilization efficiency, achieving the coordinated development of capital deepening and low-carbon transition.

This paper still has several limitations. First, the analysis of institutional mechanisms relies mainly on theoretical interpretation and indirect empirical evidence, without directly econometrically testing institutional variables such as fiscal transfers and promotion incentives; future work can refine institutional indicators and carry out quantitative identification of the mechanisms. Second, due to data availability, the effective observation window of the core fixed-asset investment data ends in 2016, which cannot fully capture the latest trends of capital deepening in recent years. Third, this paper uses total urban carbon emissions and does not distinguish production-side from consumption-side emissions, making it difficult to precisely identify the heterogeneous effects of population flows on different dimensions of emissions. Future research can further refine institutional quantitative indicators, extend the data window, and decompose the emission structure, deepening the study of institution-driven capital deepening and the green paradox.

References

- Brock, W.A., Taylor, M.S., 2010. The green Solow model. *Journal of Economic Growth* 15, 127–153.
- Du, W., Li, M., 2024. Population shrinkage and green development: Evidence from China. *Journal of Environmental Management* 351, 119832.
- Ehrlich, P.R., Holdren, J.P., 1971. Impact of population growth. *Science* 171, 1212–1217.

- Fan, C.C., 2019. Industrial relocation and regional economic development in China. *Regional Studies* 53, 993–1006.
- Fan, J., Zhou, L., Zhang, Y., Shao, S., Ma, M., 2021. How does population aging affect household carbon emissions? Evidence from Chinese urban and rural areas. *Energy Economics* 100, 105356.
- Fan, Z., Zhang, J., 2010. Fiscal decentralization, transfer payments and domestic market integration (in chinese). *Economic Research Journal* 45, 53–64.
- Kaya, Y., Yokobori, K., 1997. *Environment, energy, and economy: Strategies for sustainability*. United Nations University Press.
- Lin, B., Chen, Z., 2018. Does factor market distortion inhibit the green total factor productivity in China? *Journal of Cleaner Production* 197, 25–33.
- Lin, B., Zhu, J., 2021. Impact of population aging on carbon emissions in China. *Energy Policy* 156, 112436.
- Stockman, A.C., 1981. Anticipated inflation and the capital stock in a cash-in-advance economy. *Journal of Monetary Economics* 8, 387–393.
- Weng, Z., 2026. Super-ministry reform, grassroots fiscal sustainability and population mobility (in chinese). *Collected Essays on Finance and Economics* 42, 53–64.
- Wu, J., 2014. China's green development and new urbanization: A two-dimensional study based on the SBM-DDF model (in chinese). *Journal of Beijing Normal University (Social Sciences)* , 130–139.
- Xi, P., Liang, R., Xie, Z., Su, G., 2017. Fiscal pressure, overcapacity and supply-side reform (in chinese). *Economic Research Journal* 52, 86–102.
- Xu, X., Wang, X., Shu, Y., 2007. Local officials and economic growth: Evidence from the exchange of provincial governors and party secretaries (in chinese). *Economic Research Journal* , 18–31.
- York, R., Rosa, E.A., Dietz, T., 2003. STIRPAT, IPAT and ImPACT: Analytic tools for unpacking the driving forces of environmental impacts. *Ecological Economics* 46, 351–365.
- Zhang, N., 2014. Does urbanization improve the efficiency of China's green development? (in chinese). *Economic Review* , 38–49.
- Zhang, S., Liu, Z., 2013. Industrial upgrading and coordinated regional development: From global value chains to domestic value chains (in chinese). *Economic Management Journal* 35, 30–40.
- Zhao, B., 2025. Population change, consumption structure and low-carbon transition (in chinese). *Journal of Quantitative and Technological Economics* 42, 5–28.
- Zhou, L., 2007. A study of the promotion tournament model of Chinese local officials (in chinese). *Economic Research Journal* , 36–50.
- Zhu, Q., Peng, X., 2012. The impacts of population change on carbon emissions in China during 1978–2008. *Environmental Impact Assessment Review* 36, 1–8.